

MH1301 Discrete Mathematics

Handout 5: Linear Recurrence Relation

In this and the next handout, we focus on solving recurrence relations. For this handout, we deal with recurrence relations that are linear and homogeneous.

- Linear homogeneous recurrence relations,
- Linear homogeneous recurrence relations of degree 2.

Linear Homogeneous Recurrence

A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where $c_1, \dots, c_k$ are real numbers and $c_k \neq 0$.

- all constants

- homogeneous & linear

- degree k

Smallest subscript we see where c_k is non-zero

these are all constant

all the terms are a_{n-i} where $i = 1, 2, \dots, k$

Linear Homogeneous Recurrence

Examples

- $f_n = f_{n-1} + f_{n-2}$ ✓ $\rightarrow$ linear and homogeneous
- $a_n = a_{n-1} + a_{n-2}^2$ is not linear $\rightarrow$ homogeneous only
- $H_n = H_{n-1} + 2$ is not homogeneous $\rightarrow$ maybe linear, not sure
 $\hookrightarrow 2$ is not in the form H_{n-i} where $i=1, 2, 3, \dots$
- $X_n = n \cdot X_{n-1}$ is not, because the coefficient n is not a constant
 $\hookrightarrow$ homogeneous only

Question

How to solve the linear homogeneous recurrence relation of degree k with constant coefficients, given k initial conditions?

Solutions of Recurrence Relations are unique

Suppose we are given the initial conditions, say, $a_0, a_1, a_2, \dots, a_{k-1}$. The recurrence relation of degree k ,

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

can then be used to give the values of $a_k, a_{k+1}, \dots$

As a consequence of the Principle of Strong Mathematical Induction, the sequence (a_n) is **uniquely determined** by the initial k conditions and the recurrence relation.

↓
so there is a unique
solution

Linear Homogeneous Recurrence with degree 2

Let's forget about the initial conditions for now.

Question

Consider a special case where degree $k = 2$:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}.$$

How many solutions of this recurrence relation can you find out?

(Hint: consider $a_n = r^n$ for some value r as a start.)

Linear Homogeneous Recurrence with degree 2

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}.$$

If $a_n = r^n$ is a solution, we have

characteristic equation

$$r^n = c_1 r^{n-1} + c_2 r^{n-2}$$

factor out r^{n-2}

$$\Rightarrow r^2 - c_1 r - c_2 = 0$$

quadratic, can solve

$$\Rightarrow \text{two roots } r_1, r_2.$$

- ① $a_n = r_1^n$ is a solution.
- ② $a_n = r_2^n$ is a solution.
- ③ for any constant h , $a_n = h \cdot r_1^n$ or $h \cdot r_2^n$ is a solution.
- ④ $a_n = r_1^n + r_2^n$ is a solution.
- ⑤ for any constants h_1, h_2 , $a_n = \boxed{h_1 \cdot r_1^n + h_2 \cdot r_2^n}$ is a solution.

Linear Homogeneous Recurrence with degree 2

$$\begin{aligned}a_n &= c_1 a_{n-1} + c_2 a_{n-2} \\ \Rightarrow r^n &= c_1 r^{n-1} + c_2 r^{n-2} \\ \Rightarrow r^2 - c_1 r - c_2 &= 0 \\ \Rightarrow \text{two roots } r_1, r_2 \\ \Rightarrow a_n &= h_1 r_1^n + h_2 r_2^n.\end{aligned}$$

recurrence relation

assuming $a_n = r^n$ is a solution

characteristic equation

characteristic roots

Examples

The characteristic equation of

- $f_n = f_{n-1} + f_{n-2}$ is $r^2 - r - 1 = 0$.
- $a_n = a_{n-1} + 3a_{n-2} - 2a_{n-3}$ is $r^3 - r^2 - 3r + 2 = 0$.
- $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ is

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_{k-1} r - c_k = 0.$$


$$r^n = r^{n-1} + 3r^{n-2} - 2r^{n-3}$$

Linear Homogeneous Recurrence with degree 2

Theorem 1 (Section 8.2)

Given recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

suppose that its **characteristic equation**

$$r^2 - c_1 r - c_2 = 0$$

has two **distinct** roots r_1 and r_2 . Then the sequence (a_n) is a solution of the recurrence relation **if and only if** (1)

$$a_n = h_1 \cdot r_1^n + h_2 \cdot r_2^n \quad (\leftarrow (2))$$

for $n = 0, 1, 2, \dots$, where h_1, h_2 are constants.

Proof of Theorem 1 ($\Leftarrow$)

from previous slide

Proof (2) implies (1)

To show the “if” direction, we need to show that the sequence (a_n) with $a_n = h_1 \cdot r_1^n + h_2 \cdot r_2^n$ is a solution of the recurrence relation.

(2)

(1)

Since r_1 and r_2 are roots of $r^2 - c_1 r - c_2 = 0$, it follows that

$$r_1^2 = c_1 r_1 + c_2 \quad \text{and} \quad r_2^2 = c_1 r_2 + c_2.$$

Therefore,

$$\begin{aligned} c_1 a_{n-1} + c_2 a_{n-2} &= c_1 \overbrace{(h_1 r_1^{n-1} + h_2 r_2^{n-1})}^{a_{n-1}} + c_2 \overbrace{(h_1 r_1^{n-2} + h_2 r_2^{n-2})}^{a_{n-2}} \\ &= h_1 r_1^{n-2} (c_1 r_1 + c_2) + h_2 r_2^{n-2} (c_1 r_2 + c_2) \\ &= h_1 r_1^{n-2} r_1^2 + h_2 r_2^{n-2} r_2^2 \\ &= h_1 r_1^n + h_2 r_2^n \\ &= a_n. \end{aligned}$$

~~This~~ This shows that (a_n) with $a_n = h_1 r_1^n + h_2 r_2^n$ is a solution of the recurrence relation.

Proof of Theorem 1 ($\implies$) *start proving from (i)*

To show the “only if” direction, we need to show that every solution (a_n) of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ has the form

$$a_n = h_1 r_1^n + h_2 r_2^n \quad \text{where } h_1 \text{ and } h_2 \text{ are constants.}$$

Suppose (a_n) is a solution and the initial conditions $a_0 = C_0$ and $a_1 = C_1$ hold. Since we want them to fit the form $a_n = h_1 r_1^n + h_2 r_2^n$, let us set h_1 and h_2 to satisfy the pair of equations

$$C_0 = h_1 + h_2 \quad \text{and} \quad C_1 = h_1 r_1 + h_2 r_2 \quad \text{--- } (*)$$

$$(*) \times r_1 - (*) : \quad r_1 C_0 - C_1 = h_2 r_1 - h_2 r_2$$

Solving the pair of equations give $h_2 = \frac{r_1 C_0 - C_1}{r_1 - r_2}$

$$h_1 = \frac{C_1 - C_0 r_2}{r_1 - r_2}$$

$$h_2 = C_0 - h_1 = \frac{C_0 r_1 - C_1}{r_1 - r_2}.$$

Note that these two expressions for h_1 and h_2 require $r_1 \neq r_2$ to be valid, and that is guaranteed by the fact that the characteristic equation has two distinct roots (and so $r_1 \neq r_2$).

Next, we show via Strong Induction that a_n takes the required form.

With these two values of h_1 and h_2 , we know that a_0 and a_1 takes the form $a_n = h_1 r_1^n + h_2 r_2^n$ for $n = 0$ and 1 .

Suppose a_{n-1} and a_{n-2} both take the required form. Consider a_n .

$$\begin{aligned} a_n &= c_1 a_{n-1} + c_2 a_{n-2} \\ &= c_1 (h_1 r_1^{n-1} + h_2 r_2^{n-1}) + c_2 (h_1 r_1^{n-2} + h_2 r_2^{n-2}) \rightarrow \text{Inductive Hypothesis} \\ &= h_1 r_1^{n-2} (c_1 r_1 + c_2) + h_2 r_2^{n-2} (c_1 r_2 + c_2) \\ &= h_1 r_1^{n-2} r_1^2 + h_2 r_2^{n-2} r_2^2 \\ &= h_1 r_1^n + h_2 r_2^n. \end{aligned}$$

Strong MI

Noting that solutions are uniquely determined by the initial k conditions and the recurrence relation, we complete the “only if” proof.

Linear Homogeneous Recurrence with degree 2

Question

How to solve a Linear Homogeneous Recurrence with degree 2 **given** **initial values of** a_0, a_1 ?

Solve the linear system

$$\begin{cases} a_0 = h_1 \cdot r_1^0 + h_2 \cdot r_2^0 = h_1 + h_2, \\ a_1 = h_1 \cdot r_1^1 + h_2 \cdot r_2^1 = h_1 \cdot r_1 + h_2 \cdot r_2. \end{cases}$$

The values of h_1 and h_2 can be uniquely decided.

In fact, we solved this on slide 11.

$$h_1 = \frac{a_1 - a_0 r_2}{r_1 - r_2}$$
$$h_2 = a_0 - h_1 = \frac{a_0 r_1 - a_1}{r_1 - r_2}.$$

Example 3 (Section 8.2)

What is the solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?

Solution:

- Its characteristic equation is $r^2 - r - 2 = 0$.

The roots are $r_1 = 2$ and $r_2 = -1$.

- Hence, (a_n) is the solution of the recurrence if and only if

$$a_n = h_1 \cdot 2^n + h_2 \cdot (-1)^n.$$

- From the initial conditions, we have

$$\begin{cases} a_0 = 2 = h_1 + h_2 \\ a_1 = 7 = h_1 \cdot 2 + h_2 \cdot (-1) \end{cases} \implies \begin{cases} h_1 = 3 \\ h_2 = -1. \end{cases}$$

- Therefore the solution is $a_n = 3 \cdot 2^n - (-1)^n$.

*Once you have fixed initial conditions,
you have fixed sequence that gives
the solⁿ → degree 2
characteristic
relation*

Example 4 (Section 8.2) Fibonacci Sequence

What is the solution of the Fibonacci recurrence relation $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0, f_1 = 1$?

↓
0, 1, 1, 2, 3, 5, 8, 13

Solution:

- Its characteristic equation is $r^2 - r - 1 = 0$.

Its roots are $r_1 = \frac{1+\sqrt{5}}{2}$ and $r_2 = \frac{1-\sqrt{5}}{2}$.

- Hence, (f_n) is a solution of the recurrence if and only if

$$f_n = h_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n + h_2 \cdot \left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Example 4 (Section 8.2) Fibonacci Sequence

Solution continue:

- From the initial conditions, we have

$$\begin{cases} f_0 = 0 = h_1 + h_2 \\ f_1 = 1 = h_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right) + h_2 \cdot \left(\frac{1-\sqrt{5}}{2}\right) \end{cases} \Rightarrow \begin{cases} h_1 = \frac{1}{\sqrt{5}} \\ h_2 = -\frac{1}{\sqrt{5}} \end{cases}$$

- Therefore the solution is

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n, \quad n \geq 0$$

Linear Homogeneous Recurrence with degree 2

Theorem 2 (Section 8.2)

Given recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

suppose that its characteristic equation

$$r^2 - c_1 r - c_2 = 0$$

has only **one** root r_0 . Then the sequence (a_n) is a solution of the recurrence relation **if and only if**

$$a_n = h_1 \cdot r_0^n + h_2 \cdot n \cdot r_0^n$$

last few slides:
 $a_n = h_1 r_1^n + h_2 r_2^n$

for $n = 0, 1, 2, \dots$, where h_1, h_2 are constants.

Proof is left as an Exercise. (Ex. 8.2.8.)

Example - Plants

Each plant reproduce a new plant during first year of its life, then never again, and all plants live forever.

Assume there is 1 plant in year 1. How many plants are there in year n ?

Solution: Let p_n denote the number of plants in year n .

- Initial condition: $p_0 = 0, p_1 = 1$.
- Recurrence: p_n is composed of two parts:

$$p_n = p_{n-1} + p_{n-1} - p_{n-2}$$

- ▶ what we already had last year
- ▶ new plants produced this year

Example - Plants

Solution: So the recurrence relation is

$$p_n = 2p_{n-1} - p_{n-2}.$$

- Its characteristic equation is $r^2 - 2r + 1 = 0$.
It has only one root $r_0 = 1$.
- Hence, (p_n) is the solution of the recurrence if and only if

$$\begin{aligned} a_n &= h_1(r_0)^n + h_2 n (r_0)^n \\ p_n &= h_1 \cdot 1^n + h_2 \cdot n \cdot 1^n. \end{aligned}$$

- From the initial conditions, we have

$$\begin{cases} p_0 = 0 = h_1 \\ p_1 = 1 = h_1 + h_2 \end{cases} \implies \begin{cases} h_1 = 0 \\ h_2 = 1. \end{cases}$$

- Therefore the solution is $p_n = 1 \cdot n \cdot 1^n = n$. $n \geq 0$

Example 5 (Section 8.2)

What is the solution of the recurrence $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1, a_1 = 6$?

Solution:

- Its characteristic equation is $r^2 - 6r + 9 = 0$. Its root is $r_0 = 3$.
- Hence, (a_n) is the solution of the recurrence if and only if

$$a_n = h_1 \cdot 3^n + h_2 \cdot n \cdot 3^n.$$

- From the initial conditions, we have

$$\begin{cases} a_0 = 1 = h_1 \\ a_1 = 6 = h_1 \cdot 3 + h_2 \cdot 3 \end{cases} \implies \begin{cases} h_1 = 1 \\ h_2 = 1. \end{cases}$$

- Therefore the solution is $a_n = 3^n + n \cdot 3^n = (n+1)3^n, n \geq 0$

Linear Homogeneous Recurrence

Theorem 3 (Section 8.2)

Given recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

suppose that its characteristic equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$$

has k **distinct** roots $r_1, \dots, r_k$. Then the sequence (a_n) is a solution of the recurrence relation **if and only if**

$$a_n = h_1 \cdot r_1^n + h_2 \cdot r_2^n + \cdots + h_k \cdot r_k^n$$

for $n = 0, 1, 2, \dots$, where $h_1, \dots, h_k$ are constants.

Proof is left as an Exercise. (Ex. 8.2.12.)

Example 6 (Section 8.2)

What is the solution of the recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with $a_0 = 2, a_1 = 5, a_2 = 15$?

Solution:

- Its characteristic equation is $r^3 - 6r^2 + 11r - 6 = 0$. Its roots are $r_1 = 1, r_2 = 2, r_3 = 3$.
- Hence, (a_n) is the solution of the recurrence if and only if

$$a_n = h_1 \cdot 1^n + h_2 \cdot 2^n + h_3 \cdot 3^n.$$

- From the initial conditions, we have

$$\begin{cases} a_0 = 2 = h_1 + h_2 + h_3 \\ a_1 = 5 = h_1 + h_2 \cdot 2 + h_3 \cdot 3 \\ a_2 = 15 = h_1 + h_2 \cdot 4 + h_3 \cdot 9 \end{cases} \implies \begin{cases} h_1 = 1 \\ h_2 = -1 \\ h_3 = 2 \end{cases}$$

- Therefore the solution is $a_n = 1 - 2^n + 2 \cdot 3^n, n \geq 0$

Simultaneous recurrence relations

Solve the simultaneous recurrence relations

* just treat as normal simultaneous equations

$$a_n = a_{n-1} + b_{n-1}, \quad \text{---} \textcircled{1}$$

$$b_n = a_{n-1} - b_{n-1}, \text{ — (2)}$$

with initial values $a_0 = 1$ and $b_0 = 2$.

From (2), make a_{n-1} the subject.

$$a_{n-1} = b_n + b_{n-1} \text{ --- (3)}$$

Sub ③ into ①

$$\underbrace{b_{n+1} + b_n}_{a_n} = \underbrace{b_n + b_{n-1} + b_{n-1}}_{a_{n-1}}$$

$$\therefore b_{n+1} = 2b_n - 1$$

$$b_n - 2b_{n-2} = 0 \quad b_{n-1} \quad b_{n-2} \text{ constant}$$

Simultaneous recurrence relations

$\therefore$ Characteristic Equation : $r^2 = 2$

$$\therefore r = \pm\sqrt{2}$$

$$r_1 = \sqrt{2}, r_2 = -\sqrt{2}$$

$$b_n = h_1(\sqrt{2})^n + h_2(-\sqrt{2})^n$$

$$b_0 = 2: 2 = h_1 + h_2 \text{ ————— } \textcircled{4}$$

$$b_1 = a_0 - b_0: 1 - 2 = -1: -1 = h_1\sqrt{2} + h_2(-\sqrt{2})$$

$$\frac{-1}{\sqrt{2}} = h_1 - h_2 \text{ ————— } \textcircled{5}$$

$$\textcircled{4} + \textcircled{5}: 2h_1 = 2 \cdot \frac{1}{\sqrt{2}} \Rightarrow h_1 = 1 - \frac{1}{2\sqrt{2}}$$

$$\text{by } \textcircled{4}, h_2 = 2 - h_1 = 1 + \frac{1}{2\sqrt{2}}$$

$$\therefore b_n = \left(1 - \frac{1}{2\sqrt{2}}\right)(\sqrt{2})^n + \left(1 + \frac{1}{2\sqrt{2}}\right)(-\sqrt{2})^n$$

Solve for a_n using $\textcircled{3}$

Multiplicity (Optional)

↓
not tested

Given an equation

$$r^k - c_1 r^{k-1} - \dots - c_{k-1} r - c_k = 0$$

If

$$\begin{aligned} r^k - c_1 r^{k-1} - \dots - c_{k-1} r - c_k \\ = (r - r_1)^{m_1} \cdot (r - r_2)^{m_2} \dots (r - r_t)^{m_t} \end{aligned}$$

where $r_1, \dots, r_t$ are distinct, then we say the equation has t distinct roots $r_1, \dots, r_t$ with **multiplicities** $m_1, \dots, m_t$.

Linear Homogeneous Recurrence (Optional)

Theorem 4 (Section 8.2)

Given recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

suppose that its characteristic equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$$

has t **distinct** roots $r_1, \dots, r_t$ with **multiplicities** $m_1, \dots, m_t$. Then the sequence (a_n) is a solution of the recurrence relation **if and only if**

$$\begin{aligned} a_n = & (h_{1,0} + h_{1,1}n + \cdots + h_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (h_{2,0} + h_{2,1}n + \cdots + h_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \cdots + (h_{t,0} + h_{t,1}n + \cdots + h_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $h_{i,j}$ are constants for any i, j .

Proof is left as an Exercise. (Ex. 8.2.39.)

Example (Optional)

What is the solution of the recurrence $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with $a_0 = 1, a_1 = -2, a_2 = -1$?

Solution:

- Its characteristic equation is $r^3 + 3r^2 + 3r + 1 = 0$. Since $r^3 + 3r^2 + 3r + 1 = (r + 1)^3$, it has one root $r_1 = -1$ of multiplicity 3.
- Hence, (a_n) is the solution of the recurrence if and only if $a_n = h_{1,0} \cdot (-1)^n + h_{1,1} \cdot n \cdot (-1)^n + h_{1,2} \cdot n^2 \cdot (-1)^n$.
- From the initial conditions, we have

$$\begin{cases} a_0 = 1 = h_{1,0}, \\ a_1 = -2 = h_{1,0} - h_{1,1} - h_{1,2}, \\ a_2 = -1 = h_{1,0} + h_{1,1} \cdot 2 + h_{1,2} \cdot 4, \end{cases} \implies \begin{cases} h_{1,0} = 1, \\ h_{1,1} = 3, \\ h_{1,2} = -2. \end{cases}$$

- Therefore the solution is $a_n = (-1)^n + 3n \cdot (-1)^n - 2n^2 \cdot (-1)^n$.

Reading Assignment

Rosen (7th edition):

- Chapter 8.1, Applications of Recurrence Relations
- Chapter 8.2, Solving Linear Recurrence Relations